

A falsification protocol for simulated embodied spiking substrates: Separating substrate-carried information from readout-channel information

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Abstract

When a population of spiking neurons is read out to control a simulated body, “the substrate computes” is often claimed from a decoded error or a mutual-information value. We argue that such a claim is not falsifiable unless the substrate is swapped for a *matched dead substrate* that preserves the single-neuron marginal firing statistics (mean, variance, silent fraction) while being statistically independent of the task, and unless the readout is itself controlled. We build a minimal simulated embodied hub (2,000 neurons, fixed sparse wiring, one calibrated decision map) and run a battery of gates: (B) substrate swap under a fixed readout; (B2) a *paired population null suite* (per-neuron circular shift, block shuffle, identity swap) with a false-positive control and a fixed-weight readout-invariance test; (C) readout-channel ablation; (D) cross-validated ridge readout of the *same* spikes; (D bis) a small nonlinear (MLP) readout on the same spikes; (E) memory-demanding tasks; and (F) a recurrence ablation. The matched

dead substrate is task-independent by construction and stays at the null floor under every readout (one-sided $p = 1/64$ per profile, 6/6 seeds; under the paired null suite, 54/54 task-varying cells at $p \leq 0.05$ for each of the three null types), with zero false positives in the dead-drive control (0/18) and a fixed-weight readout-invariance result that survives at the resolution of the null sample (12/12, $p = 0.0476$). The living substrates do not uniformly survive. LIF carries the task through a cross-validated linear readout ($\Delta\text{RMSE}_{\text{floor}} = +0.094$ vs. the mean-predictor floor; MI significant in 24/30 runs) and a small nonlinear readout does not change that materially (+0.105); a seed-level paired test at $n = 20$ seeds is decisive for LIF (20/20 seeds same direction, $p \approx 10^{-5}$). The apparent Izhikevich tracking seen under the fixed calibrated readout is largely a *readout-channel artifact*: randomizing the readout projection collapses both living substrates to the dead floor (3/3 seeds), a cross-validated linear readout of the same IZH spikes leaves RMSE essentially at the floor ($\Delta\text{RMSE}_{\text{floor}} = -0.008$), and a nonlinear readout recovers only a small amount (+0.012, 16/20 seeds same direction). None of the substrates shows linearly decodable history in the present-window population state (Gate E negative), so memory claims are not certified here. The ranking is not architecture-bound: a second hub ($N = 4000$, $p_{\text{re}} = 0.02$) reproduces it (LIF +0.104 vs. IZH +0.004 over the dead null). We conclude: (i) matched dead-substrate controls for the *material* of a substrate are cheap, decisive, and should precede scoreboard comparisons; (ii) readout *robustness* is a stronger falsification test than a single scoreboard, and we reserve the word “invariance” for what our experiments cannot show; and (iii) open-loop diagnosis cannot certify memory, which must be demonstrated in the closed loop.

1 Introduction

Physical and simulated substrates for neural computation—in-vitro cultures (Kagan et al., 2022), neuromorphic silicon (Maass, 1997; Lukosevicius Jaeger, 2009; Wringe et al., 2025), or toy spiking hubs—are routinely evaluated by coupling them to a task and reporting a score: a decoded error, a correlation, a bit rate. Two distinct claims are usually conflated under that score. The first is *substrate-carried information*: spike patterns produced by the substrate contain task-relevant statistics. The second is *readout-channel information*: the fixed mapping used to translate spikes into a decision happens to recover (or inject) task statistics. A scoreboard alone cannot tell them apart, and the recent discussion around “intelligence in a dish” (Kagan et al., 2022; Balci et al., 2023; Kagan et al., 2023; Smirnova et al., 2023) shows that this conflation has consequences far beyond toy models.

The contribution of this note is a *falsification protocol* that separates the two. It rests on three ingredients, all cheap:

- **A matched dead substrate** (Gate B). The wiring, bias, drive and readout are kept fixed; only the per-neuron dynamics change. The dead substrate fires Poisson processes with per-neuron rates matched to the LIF marginal histogram, but is statistically independent of the task. Any substrate that cannot beat this null under a fixed readout is not falsifiably carrying the task.
- **A paired population null suite** (Gate B2). A null built from a single i.i.d. Poisson process matches only the single-neuron marginal and discards temporal structure; a stronger, still cheap control is to build *sample-matched*

surrogates of the substrate’s own spike trains (per-neuron circular shift, block shuffle, identity swap) and test the observed statistic against them (Theiler et al., 1992). We also add a dead-drive false-positive control and a fixed-weight readout-invariance test.

- **Readout controls** (Gates C, D, D bis and F). Gate C randomizes the readout projection into the fixed decision map; Gate D replaces the hand-tuned readout with a cross-validated ridge readout trained on the *same* spikes; Gate D bis adds a small nonlinear (MLP) residual readout to bound how much linearity hides; Gate F ablates recurrence. Task information that remains recoverable after *all* controls is evidence the substrate activity itself, rather than the calibrated interface alone, carries the signal. Throughout we reserve stronger language (“invariance”) for what the experiments cannot show, and use *readout robustness* for what they do.

Gate E then asks the harder question—whether present-window population activity carries task *history*—which is the minimum a memory claim needs. Throughout, the replication unit is the seed. Gates B/D/E pool 6 seeds $\times$ 5 profiles; Gate C uses 3 seeds; Gate D is additionally repeated at $n = 20$ *seeds* with an exact seed-level paired permutation test; Gates B2 uses the 6 seeds $\times$ 10 profiles with $K = 50$ paired nulls per cell.

2 Related work

Six literatures touch the same question from different angles.

- (1) **Embodied neural computation.** “Intelligence without representation” (Brooks, 1991) and morphology-computation continua (Pfeifer Bongard, 2006)

established that behavior is jointly determined by body, wiring and control; more recent work formalizes that for neuromorphic-body systems (D’Angelo et al., 2026; Bartolozzi et al., 2022). These works score systems, but do not isolate the substrate’s material from its interface. Our hub is deliberately a *simulated* body; the protocol is what transfers, not the embodiment.

(2) In-vitro substrates and the DishBrain debate. Kagan et al. reported task-relevant responses of cultured cortical neurons in a closed loop (Kagan et al., 2022); the follow-up correspondence (Balci et al., 2023; Kagan et al., 2023) turned precisely on whether the observed signal is attributable to the biology or to the measurement/readout interface, and the broader organoid-intelligence program (Smirnova et al., 2023; Zhou et al., 2026) inherits the same methodological question. Our protocol is a minimal, fully deterministic enactment of that critique.

(3) Spiking and reservoir computing. Spiking neuron models (Maass, 1997; Izhikevich, 2003; Izhikevich, 2004; Gerstner & Kistler, 2002) and liquid state machines / echo-state networks (Maass et al., 2002; Jaeger, 2001; Lukosevicius & Jaeger, 2009) show that random recurrent populations can support surprisingly capable online regression. The standard evaluation there is already readout-shaped (train a readout, report performance); recent critiques of reservoir-computing benchmarks (Wringe et al., 2025; Yan et al., 2024) make explicit that task scores alone are hard to compare and easy to over-read. What is usually missing is the *dead-material control*—the same readout trained on matched independent Poisson spikes (Lukosevicius & Jaeger, 2009)—and paired sample-matched surrogates of the substrate’s own activity.

(4) Neuromorphic benchmarks. NeuroBench (Yik et al., 2025) and related efforts define task suites, metrics and complexity accounting for neuromorphic algorithms and systems, and embodied extensions are emerging (D’Angelo et al., 2026). Benchmark suites standardize *what* to score; they do not, by themselves, supply the substrate-*material* null that our protocol requires.

(5) Population decoding and information measures. Neural coding theory provides the tools: population decoding with regularized linear readouts (Georgopoulos et al., 1986; Quiroga Panzeri, 2009; Paninski et al., 2004; Hoerl Kenard, 1970), mutual information and its bias correction (Shannon, 1948; Kraskov et al., 2004; Panzeri et al., 2007), transfer entropy with surrogate-based significance (Schreiber, 2000; Walker Newhall, 2018), and nonlinear readouts (kernel methods, MLPs, population models) (Paninski et al., 2004; Steinmetz et al., 2019). Our Gate D uses exactly this machinery; the dead-substrate null is the novel anchor against which those measures should be read.

(6) Null models and statistical rigor. Surrogate-data methods (Theiler et al., 1992), permutation statistics (Ernst, 2004), pseudo-replication warnings (Hurlbert, 1984), and closed-loop validation practice in brain-machine interfaces (Carmenta et al., 2003; Nicolelis, 2003) all argue that a structurally matched null is the difference between an effect and an artifact. Gates B2 and C are surrogate tests applied to the *substrate and readout channel* rather than to a single scalar series.

3 Methods

3.1 Embodied hub and wiring

$N = 2,000$ neurons, $N_{\text{win}} = 200$ control windows of $C_W = 20$ integration steps (10 ms bins, 0.5 ms integration). The input projection $\mathbf{g} \in \mathbb{R}^{N \times 1}$ is standard-normal; recurrence $\mathbf{W}$ is sparse with connection probability $p_{\text{re}} = 0.01$ and entries drawn $\mathcal{N}(0, 1/\sqrt{Np_{\text{re}}})$ (variance-normalized so the network operates near stability); the readout projection is $\mathbf{w}_r = \mathbf{g}$ (channel aligned with the input projection), with a per-substrate scale k_{ro} (below). The decision map is the calibrated one of the reference system (Zhou et al., 2026), $\hat{x} = \text{clip}((\theta_{\text{sig}} - 0.0615)/0.277, 0, 0.75)$, clamped to 0.75 m/s. Five control profiles (**baseline**, **multi_step**, **sine**, **descend**, **pulse**), six seeds $\{1, 7, 13, 29, 55, 91\}$ for Gates B/C/D/E, and Gate B2 additionally uses five profiles of a broader battery (**ramp**, **perturb**, **step_pulse**, **chirp**, **noise**). Gate D is repeated with $n = 20$ seeds (Section 4.9).

3.2 Substrates and the matched dead null

Three dynamics on identical wiring, bias, drive and readout constants:

- **LIF**: leaky integrate-and-fire, $\tau = 20$ ms, threshold $v_{\text{th}} = 1$, reset 0, refractory 2 ms.
- **Izhikevich**: regular spiking parameters [“RS”, (0.02, 0.2, -65 , 8)] integrated with the 0.5 ms step; adaptation dimension u and $a = 0.02$, $b = 0.2$, $c = -65$, $d = 8$.
- **Matched dead null (Poisson)**: per-neuron events drawn independently each integration step from Bernoulli trials with probability r_i ; the r_i are esti-

mated from LIF’s per-neuron firing in a constant-input ($u = 0.5$) calibration run (Theiler et al., 1992), so the marginal per-neuron rates match. Table 1 reports the match quality explicitly; the null is *statistically independent* of u by construction.

The key design point is that the match is at the level of the *single-neuron marginal rate histogram*—mean μ_r , variance, and the fraction of near-silent neurons—not at the level of any temporal or population structure. This is what makes Gate B a *substrate* test rather than a firing-rate test. Table 1 (from `gate_b_summary.json`) gives $\mu_r = 8.09$ Hz, SD 15.30 Hz, Fano $\sigma^2/\mu = 28.9$ (overdispersed vs. a homogeneous Poisson) and 74% neurons with zero estimated rate.

Table 1: Match quality of the dead null to the LIF marginal: per-neuron mean rate, SD, Fano factor σ^2/μ , and the silent fraction (neuron with $\hat{r}_i \approx 0$). The null draws Bernoulli events at the matched per-neuron probabilities, independent of u .

statistic	LIF marginal (r_i)	matched IID Poisson	$N = 2000$ neurons	Fano
per-neuron mean rate (Hz)	8.091	8.091	both	–
per-neuron StdDev (Hz)	15.296	$\sqrt{\mu}$ only	–	28.916
fraction silent ($r_i = 0$)	0.740	$e^{-\mu}$ expected	1480/2000	–

3.3 Calibration without leakage

k_{ro} is fixed per substrate from a *constant-input* calibration run ($u = 0.5$, seed 7): $k_{\text{ro}} = 0.2/\langle \mathbf{w}_r \cdot \text{rate} \rangle$ sets the decision series at the middle of the map’s active range. Crucially, calibration never sees a *task-varying* input, so no task information leaks

into the readout constant. The same k_{ro} survives Gate C: the randomized projection is rescaled to the same norm ($\|\mathbf{w}_r^{\text{rand}}\| = \|\mathbf{g}\|$), so the constant remains comparable. The Poisson null uses the LIF-derived k_{ro} plus a $k_{\text{ro}}^{\text{poisson}}$ from its own constant-input run, again $0.2/\langle\cdot\rangle$.

3.4 Gate B2: paired population null suite

Gate B’s dead null is i.i.d. and therefore destroys any temporal structure. To test the substrate against its own activity, for each profile and seed we build $K = 50$ sample-matched surrogates of the recorded counts and recompute the same readout statistic:

- **circshift**—per-neuron circular shift of the window sequence (preserves each neuron’s marginal and autocorrelation on the torus);
- **blockshuffle**—shuffle contiguous blocks of windows (block length $B = 5$), destroying long-range task alignment while keeping short-range structure;
- **identityswap**—exchange the window order across neurons (preserves the exact per-window population envelope, destroying neuron–task alignment).

The observed statistic is compared to the K surrogate values with a one-sided rank test; a cell is significant at $\alpha = 0.05$. As a false-positive control we rerun the whole pipeline with `drive_gain=0` (no task signal) on three profiles (**sine**, **noise**, **ramp**) $\times 6$ seeds $\times 2$ substrates: no cell may be significant. Finally, a *fixed-weight readout-invariance* test refits nothing: the ridge weights learned on the real spikes are frozen and applied unchanged to real and surrogate counts, so the test measures whether recoverable information survives the surrogate and not whether a fresh model can adapt to it.

3.5 Gate C: readout-channel control

For each seed s , draw an independent zero-mean normal vector $\mathbf{w}_s^{\text{rand}}$, center it, and rescale to $\|\mathbf{g}\|$ (RNG seed 4242). Keep wiring, bias, drive and the fixed decision map; only the readout projection changes. If apparent tracking under the aligned channel survives randomization, it is substrate-independent; if it collapses to the dead floor, the apparent tracking depended on the readout alignment (whether the substrate activity *also* contains readout-recoverable task information is asked separately by Gate D).

3.6 Decoders (Gates D and D bis)

On the *same* recorded spike trains:

1. **canon**—the fixed calibrated channel $k_{\text{ro}} \mathbf{w}_r \cdot (\text{counts}/C_W) + \text{decide}$ (Gate B reference);
2. **ridge**—cross-validated L2-regularized linear readout of population counts with 13 lags on a log-grid in $[10^{-5}, 10]$, fit on interleaved train/val windows (every phase appears in every fold) solved via the Woodbury identity and verified against a direct solve ($\max |\Delta| < 4 \times 10^{-10}$; `verify_ridge.py`);
3. **meanpd**—constant-predictor floor (var-around-run-mean);
4. **passthru**— $0.75 u$ directly (RMSE = 0 by construction, the no-substrate ceiling);
5. **mlp** (Gate D bis)—the fitted ridge readout plus a small one-hidden-layer tanh network ($2000 \rightarrow 32 \rightarrow 1$, L2, gradient clipping) trained on the ridge’s train residual, with its own penalty selected on the validation split. This *linear-skip residual* form starts exactly at the optimal linear readout, so

a nonlinear gain is possible only if it generalizes on held-out windows; a larger net overfits (the split has only ≈ 66 train windows), which is itself a data-budget observation for a hub of this size.

We report the *RMSE improvement over the mean-predictor floor*,

$$\Delta\text{RMSE}_{\text{floor}} = \text{RMSE}_{\text{meanpd}} - \text{RMSE}_{\text{decoder}},$$

taking the constant predictor as the zero-information baseline: positive values mean the decoder beats a constant. We deliberately avoid an information-unit name for $\Delta\text{RMSE}_{\text{floor}}$ because $-\text{RMSE}$ is a loss, not an entropy; the reader should not read it as a bit or nat quantity. When comparing a living substrate to the dead null we use the difference-in-differences $\Delta\text{RMSE}_{\text{floor}}(\text{living}) - \Delta\text{RMSE}_{\text{floor}}(\text{null})$ so that any overfitting of the same decoder on independent spikes cancels.

3.7 Statistics and pseudo-replication

The unit of replication is the *seed* (wiring + initial state). Gate B/E pool 6 seeds; Gate D pools 6 seeds $\times$ 5 profiles = 30 runs, and is repeated at $n = 20$ seeds. We report both per-seed outcomes (e.g. “significant MI in 24/30 runs”) and pooled means, and we flag seed-level counts rather than inflating n . This follows standard pseudo-replication guidance (Hurlbert, 1984): the 200 windows within a run are not independent evidence. The seed-level test is an exact paired sign-permutation over the 20 seeds (Section 4.9); with $n = 20$ its one-sided resolution floor is $\approx 10^{-5}$ at 10^5 permutations.

3.8 Eager statistics for Gate B

For the smallest n , we report the exact paired permutation over the 6 seeds: a one-sided $p = 1/64 = 0.0156$ and two-sided $p = 2/64 = 0.03125$, with 6/6 seeds in the same direction. We state the resolution explicitly rather than quoting deeper p -values than the design supports.

Experimental design at a glance.

Table 2: The gates: what is changed, what is held fixed, and the question each gate answers. Gate F (recurrence ablation) was a remark in v1 and is promoted here to a gate.

Gate	Manipulation	Held fixed	Question
B (substrate swap)	dynamics $\rightarrow$ Poisson null with matched marginals	wiring, bias, drive, readout $\mathbf{w}_r = \mathbf{g}$, decision map	Does a task-independent dead substrate reproduce the score?
B2 (paired null suite)	counts $\rightarrow$ sample-matched surrogate (circshift / blockshuffle / identityswap), $K = 50$	wiring, drive, decision map; plus a dead-drive FP control and a fixed-weight readout-invariance test	Does the score survive surrogates of the substrate’s <i>own</i> activity, and is the test well-calibrated?
C (channel ablation)	readout projection $\rightarrow$ per-seed random $\mathbf{w}_r^{\text{rand}}$	wiring, drive, spikes, decision map	Does the apparent tracking survive breaking the aligned channel?
D (readout re-learn)	decoder $\rightarrow$ cross-validated ridge on the <i>same</i> spikes	spikes, task	Do the spike trains themselves contain readout-recoverable task information?
D bis (non-linear readout)	ridge + small tanh MLP residual on the same spikes	spikes, task	How much does the linear readout hide?
E (memory)	task $\rightarrow$ history-dependent (<code>lag</code> , <code>integ</code> , <code>stair</code>); decoder on present-window state	spikes, dynamics, readout	Is task <i>history</i> linearly decodable from the present population state?
F (recurrence)	recurrence $\mathbf{W} \rightarrow 0$ (no hub), same readout	drive, readout, decision map	Does the tested recurrence contribute at all?
G (architecture)	hub $\rightarrow$ architecture B ($N = 4000$, $p_{\text{re}} = 0.02$), same substrates & protocol	substrates, profiles, seeds, decoders, decision map	Is the substrate ranking bound to the one hub size/density?

Gates B/B2/C/D/D bis/F decompose one score into (substrate material) vs. (interface); Gate E sets the bar any memory claim must clear; Gate G checks the ranking is not architecture-bound. All are run on the *same* seeds and profiles (Gate G on a second hub).

4 Results

4.1 Gate B: substrate swap under a fixed readout

Table 3: Gate B: per-profile $\widehat{\text{RMSE}}$, MI (nats), ρ and TE-significance under the fixed calibrated readout (5 profiles, 6 seeds). “msd” is the across-seed mean.

profile	substrate	$\widehat{\text{RMSE}}$	MI	ρ	TE	p_{TE}
baseline	LIF	0.238	0.000	0.000	5e-04	0.253
baseline	Izhikevich	0.254	0.000	0.000	4e-05	0.400
baseline	Poisson (dead null)	0.399	0.000	0.000	0e+00	0.833
descend	LIF	0.167	1.034	0.933	0e+00	0.027
descend	Izhikevich	0.196	0.627	0.892	0e+00	0.000
descend	Poisson (dead null)	0.382	0.002	-0.006	0e+00	0.833
multi_step	LIF	0.131	0.681	0.994	0e+00	0.000
multi_step	Izhikevich	0.215	0.444	0.822	0e+00	0.000
multi_step	Poisson (dead null)	0.432	0.000	0.012	0e+00	0.833
pulse	LIF	0.124	0.574	0.958	0e+00	0.000
pulse	Izhikevich	0.294	0.241	0.570	0e+00	0.000
pulse	Poisson (dead null)	0.372	0.000	0.004	0e+00	0.833
sine	LIF	0.179	0.966	0.938	0e+00	0.027
sine	Izhikevich	0.236	0.614	0.879	0e+00	0.000
sine	Poisson (dead null)	0.375	0.000	-0.012	0e+00	0.833

Under the fixed calibrated readout, LIF and IZH both beat the matched dead null on every task-varying profile (Table 3). The dead null stays at $\text{MI} \approx 0$ and

$\rho \approx 0$; its TE is zero under the trial-shuffle estimator at every profile and seed. On **baseline** (constant input) *no* substrate carries information, as expected. Taken alone, Gate B would read as “the neural substrates compute”—this is precisely the reading that Gates B2, C and D correct.

4.2 Gate B2: paired null suite, calibration and invariance

Table 4: Gate B2: per substrate and surrogate type, the number of task-varying cells (profiles $\times$ seeds) with surrogate $p \leq 0.05$ out of 54, at $K = 50$ paired surrogates per cell. The identity-swap null preserves the exact per-window population envelope.

null type	substrate	sig MI cells	total	frac.
<code>circshift</code>	LIF	54	54	1.00
<code>circshift</code>	Izhikevich	54	54	1.00
<code>blockshuffle</code>	LIF	54	54	1.00
<code>blockshuffle</code>	Izhikevich	54	54	1.00
<code>identityswap</code>	LIF	54	54	1.00
<code>identityswap</code>	Izhikevich	54	54	1.00

The observed MI is above its own sample-matched surrogates in 54/54 task-varying cells for *each* of the three null types and *both* living substrates (Table 4), so the effect is not an artifact of the i.i.d. Poisson assumption. The identity-swap surrogate is the sharpest check: it keeps every window’s population counts exactly and only destroys the neuron-to-task alignment, yet the MI still collapses. The dead-drive false-positive control (`drive_gain=0`, no task signal, three profiles $\times$

six seeds $\times$ two substrates) yields *zero* significant cells, so the surrogate test is well-calibrated at this sample size. Finally, freezing the ridge weights learned on the real spikes and applying them unchanged to the surrogates leaves the task MI above the surrogate distribution in 12/12 seed–substrate cells at the resolution of the null sample ($p = 0.0476$); because nothing is refit, this is a statement about surviving information rather than about model adaptability. The three criteria—null suite, false-positive control, fixed-weight invariance—are all met (verdict PASS).

4.3 Gate C: does the channel carry it?

Table 5: Gate C: canonical RMSE under the aligned channel ($\mathbf{w}_r = \mathbf{g}$) vs. per-seed random readouts. “ d ” is $\text{RMSE}_{\text{random}} - \text{RMSE}_{\text{aligned}}$; “paired” is the fraction of seeds with $d > 0$. Both living substrates collapse (3/3 seeds); the matched dead null is unaffected.

profile	substrate	$\widehat{\text{RMSE}}_{\text{aligned}}$	$\widehat{\text{RMSE}}_{\text{random}}$	Δ	p_{paired}
sine	LIF	0.183	0.407	+0.224	3/3
sine	Izhikevich	0.235	0.396	+0.161	3/3
sine	Poisson (dead null)	0.407	0.407	+0.000	0/3
pulse	LIF	0.124	0.397	+0.273	3/3
pulse	Izhikevich	0.296	0.361	+0.065	3/3
pulse	Poisson (dead null)	0.397	0.397	+0.000	0/3

Randomizing the readout projection collapses *both* living substrates toward the dead floor: mean RMSE jumps from 0.12—0.30 (aligned) to 0.36—0.41 (random-

ized), in all three evaluated seeds per profile (Table 5). The matched dead null is unaffected ($\Delta = 0.000$, 0/3 seeds). Gate C therefore shows that performance under the fixed decision map *depends critically on the alignment of the readout projection* with the input projection. It does not by itself say the substrate is empty: a poor channel could also fail to expose genuinely present substrate information. Gate D settles that direction—whether the underlying spike trains retain readout-recoverable task information—by re-learning the readout on the same spikes.

4.4 Gate D: the readout decides who looks like it computes

Table 6: Gate D: same spikes, decoder sweep (pooled 5 profiles $\times$ 6 seeds). $\Delta\text{RMSE}_{\text{floor}} = \text{RMSE}_{\text{meanpd}} - \text{RMSE}_{\text{decoder}}$ is the RMSE improvement over the mean-predictor (no-information) floor; MI in nats; MI_{sig} is the fraction of runs with significant MI.

substrate	decoder	$\widehat{\text{RMSE}}$	$\Delta\text{RMSE}_{\text{floor}}$	MI (nats)	ρ	MI_{sig}
LIF	fixed calibrated channel	0.168	-0.018	0.651	0.765	0.800
LIF	ridge (cross-validated)	0.055	0.094	0.391	0.733	0.800
Izhikevich	fixed calibrated channel	0.239	-0.090	0.385	0.633	0.800
Izhikevich	ridge (cross-validated)	0.157	-0.008	0.250	0.458	0.767
Poisson (dead null)	fixed calibrated channel	0.392	-0.243	0.000	-0.000	0.000
Poisson (dead null)	ridge (cross-validated)	0.158	-0.009	0.018	0.038	0.033

Table 6: a cross-validated ridge readout of *the same spikes* splits the substrates:

- **LIF**: RMSE drops to 0.055—the fixed channel had left information on the ta-

ble (canon-to-ridge gap +0.113)—and the spikes carry the task: $\Delta\text{RMSE}_{\text{floor}} = +0.094$, MI significant in 24/30 runs.

- **Izhikevich:** the fixed-channel “tracking” does *not* persist: ridge collapses to 0.157, $\Delta\text{RMSE}_{\text{floor}} \approx 0$ (−0.008), MI significant in 23/30 runs but at about a quarter of the LIF rate. The Gate B IZH gap was a readout-channel artifact (confirmed by Gate C).
- **Dead null:** ridge $0.158 \approx \text{floor}$, $\text{MI} \approx 0$ (frac. 0.03)—the matched null remains at the floor under the cross-validated ridge readout (Hoerl Kennard, 1970; Paninski et al., 2004).

4.5 Gate D bis: does a nonlinear readout change the verdict?

Table 7: Gate D bis: same spikes and splits as Gate D, with a small MLP residual readout on top of the cross-validated ridge. $\Delta\text{RMSE}_{\text{floor}}$ is the improvement over the mean-predictor floor; “MI-sig” is the fraction of runs with significant MI (30 runs).

substrate	$\Delta\text{RMSE}_{\text{floor}}$ linear	$\Delta\text{RMSE}_{\text{floor}}$ MLP	MI-sig linear	MI-sig MLP
LIF	0.094	0.105	0.800	0.800
Izhikevich	-0.008	0.012	0.767	0.800
Poisson (dead null)	-0.009	-0.009	0.033	0.033

A reviewer objection to linear readouts is that the best linear decoder may hide task information that a nonlinear one would recover. We test this directly: the MLP starts from the optimal linear readout and adds a regularized nonlinear cor-

rection. Table 7 shows the verdict is essentially unchanged. LIF’s $\Delta\text{RMSE}_{\text{floor}}$ moves only from +0.094 to +0.105 (relative to the dead null, +0.103 to +0.114): the nonlinearity is neither needed nor harmful. IZH gains a small amount (from ≈ 0 to +0.012 vs. the null +0.021), still far below LIF and consistent with the interpretation that IZH’s fixed-channel tracking is mostly a readout-channel artifact. The dead null stays at the floor ($\Delta\text{RMSE}_{\text{floor}} = -0.009$; significant MI in 1/30 runs). We therefore report the linear readout as the conservative comparison and treat the MLP as a bound, not as a new positive claim.

4.6 Gate E: is there memory?

Table 8: Gate E: RMSE improvement over the mean-predictor floor, $\Delta\text{RMSE}_{\text{floor}}$ (ridge minus floor), per profile, plus the input-only 8-tap FIR reference (positive *floor* values).

profile	LIF	Izhikevich	Poisson	input-FIR8
lag	-0.139	-0.055	-0.016	0.152
integ	-0.114	-0.104	-0.012	0.144
stair	-0.141	-0.025	0.010	-0.016

Negative result. Three history-dependent tasks (**lag**: 4-window delayed reference; **integ**: running integral over 8 windows; **stair**: staircase-level switch) decoded by the ridge on present-window spikes vs. an 8-tap FIR over the input only. $\Delta\text{RMSE}_{\text{floor}}$ is negative for every substrate (LIF -0.13 , IZH -0.06 , null -0.006), while the input-FIR solves the linearly recoverable ones (+0.15). Conclusion: no substrate shows *linearly decodable task history* in the present-window population

state. Memory is not certified here; functional memory must be demonstrated in the closed loop, where previous activity feeds back into the input (Carmenta et al., 2003; Nicolelis, 2003).

4.7 Gate F: recurrence ablation

Section 4.4 reported that dropping recurrence (w ignored) leaves the canonical RMSE unchanged. We promote this to a gate. At $n = 6$ seeds, nohub and canon agree for LIF to $< 10^{-4}$ (0.1678 vs. 0.1678) and for IZH to $\approx 10^{-3}$ (0.2388 vs. 0.2390); at $n = 20$ seeds the pooled gap is -3.3×10^{-5} (LIF) and $+3.5 \times 10^{-4}$ (IZH), i.e. below the resolution of the measurement. Under the wiring and drive used here, the tested recurrence contributes negligibly to the measured canonical RMSE, so none of the above is a recurrence artifact. This is a statement about *this* hub at *this* operating point, not a general claim that recurrence is computationally inert.

4.8 Architecture robustness of the ranking

Table 9: Same Gate D battery (five profiles, six seeds, identical substrates, decoders and decision map) on a second hub, architecture B ($N = 4000$, $p_{\text{re}} = 0.02$, variance-normalized wiring), versus architecture A of Sections 3.1–4.7. $\Delta\text{RMSE}_{\text{floor}}$ is the improvement over the mean-predictor floor; “info–null” subtracts the matched dead-null floor.

architecture	substrate	$\Delta\text{RMSE}_{\text{floor}}$	info–null	MI-sig
A ($N=2000$, $p_{\text{re}}=0.01$)	LIF	0.094	0.103	0.800
A ($N=2000$, $p_{\text{re}}=0.01$)	Izhikevich	-0.008	0.001	0.767
B ($N=4000$, $p_{\text{re}}=0.02$)	LIF	0.101	0.104	0.800
B ($N=4000$, $p_{\text{re}}=0.02$)	Izhikevich	0.000	0.004	0.800

The LIF-vs-IZH gap could be an artifact of the one hub we happened to build. To test this, we rerun the *entire* Gate D battery on a different architecture—larger and denser ($N = 4000$, $p_{\text{re}} = 0.02$), with the same variance-normalized wiring so the per-neuron operating point stays comparable—while holding substrates, profiles, seeds, decoders and the decision map fixed. The ranking reproduces (Table 9): LIF’s carried information over the dead null is +0.104 (vs. +0.103 in architecture A) and IZH’s is +0.004 (vs. +0.001), both far below LIF, with the dead null at the floor (−0.004) and the same three criteria met (LIF real, IZH weak, null dead). The pooled canonical RMSE is likewise stable (LIF 0.165 vs. 0.168; IZH 0.242 vs. 0.239). Within the range of hubs tested, then, the conclusion “LIF carries linearly recoverable task information; IZH does not” is not bound to a specific hub size or density.

4.9 Seed-level statistics at $n = 20$

Table 10: Seed-level paired test at $n = 20$ seeds. For each seed we pool the gain-in- $\Delta\text{RMSE}_{\text{floor}}$ of the living substrate over the dead null across the five profiles, then run an exact paired sign-permutation over the 20 seeds (10^5 permutations).

substrate	Δ (vs null)	SE	paired perm p	seeds same dir
LIF	0.103	0.001	1e-05	20/20
Izhikevich	0.008	0.003	0.00356	16/20

To address the small- n concern directly, we reran Gate D at $n = 20$ seeds (the six canonical seeds plus fourteen new ones) with an exact seed-level paired permutation test (Table 10). LIF is decisive: the mean effect over the null is +0.103 with 20/20 seeds in the same direction ($p \approx 10^{-5}$, the permutation resolution floor). IZH is qualitatively different: a small +0.008 mean effect, 16/20 seeds in the same direction ($p \approx 3.6 \times 10^{-3}$). The pooled canonical RMSE at $n = 20$ is essentially identical to the $n = 6$ battery (LIF 0.167 vs. 0.168; IZH 0.239 vs. 0.239; ridge LIF 0.058 vs. 0.055; ridge IZH 0.153 vs. 0.158), and the dead null stays dead (significant MI in 5% of runs). We therefore report both samplings and do not inflate the effective sample size.

4.10 Verification

Four independent checks confirm the results are not implementation artifacts:

- `verify_ridge.py`—Woodbury ridge equals a direct least-squares solve (max $|\Delta| < 4 \times 10^{-10}$): Gate D numbers are the solver’s, not the optimization’s;

- `verify_null.py`—matched dead null on two out-of-family seeds {103, 201}:
MI = 0, TE = 0, $\rho = 0$ exactly;
- `verify_wr.py`—per-seed independent random readouts ($\|\mathbf{w}^{\text{rand}}\| = \|\mathbf{g}\|$,
RNG 4242): the randomization used in Gate C, verified against the aligned
baseline and the null;
- `verify_paper_numbers.py`—every number quoted in this v2 is re-derived
from the JSON artifacts and the build fails on any mismatch (see Section 6).

5 Discussion

5.1 The matched dead substrate is the falsification anchor

A scoreboard is necessary but not sufficient. Every number in Gate B that looks like “computation” is preceded by the two-line question: *does an independent Poisson process with the same marginal rates do it too?* When it does not, the material deserves a closer look; when it does, nothing follows about the substrate. Gate B2 sharpens this with surrogate tests built from the substrate’s own spike train: the effect survives samples that preserve its marginals, its short-range structure, and even its per-window population envelope, but not their task alignment. The matched null is cheap (a rate estimate plus a surrogate generator) and it should precede scoreboard comparisons in the embodied-AI literature (Kagan et al., 2022; Balci et al., 2023; Lukosevicius Jaeger, 2009; Wringe et al., 2025).

5.2 Readout robustness, not invariance, is the substantive test

Gate C makes the channel visible by breaking it: both living substrates collapse when $\mathbf{w}_r$ stops matching $\mathbf{g}$. Gate D then asks whether a readout *trained* on the same spikes can recover the task: LIF survives ($\Delta\text{RMSE}_{\text{floor}} = +0.094$), IZH does not ($\Delta\text{RMSE}_{\text{floor}} = -0.008$). Gate D bis shows a nonlinear readout does not change the picture (LIF $+0.105$, IZH $+0.012$). Readout choice is therefore a substantive modeling decision, not a detail. Claims like “this material computes ϕ ” must be re-read as “this material’s spike statistics, *via* some channel, contain readout-recoverable task-relevant structure,” and the channel must be shown to be redundant—or the claim collapses to a statement about the interface, exactly the misreading flagged in the DishBrain correspondence (Balci et al., 2023). We use *robustness* rather than *invariance* because our readouts are still a family of linear (and one small nonlinear) models; nothing here shows invariance to arbitrary readouts.

5.3 What we are not claiming

The measured canonical RMSE is unaffected by dropping the recurrent connections (Gate F); the tested recurrence contributes negligibly to the results, and Gate E is negative, so no statement here concerns memory. None of the numbers is a closed-loop result. A closed-loop falsification battery is left to future work (§5.5); this note’s protocol is the *open-loop instrument* that any closed-loop claim should first pass. Only LIF of the three materials survives the readout controls as a carrier of task information, and only for the rate-code tasks.

5.4 Limits

(i) The linear readout is now complemented by a small nonlinear readout (Gate D bis), but not by a large network or a population model; our nonlinear bound is deliberately weak, and larger decoders would need more data than the ≈ 66 windows per split that the hub provides. (ii) The matched null is *single-neuron* marginal for Gate B and sample-matched for Gate B2; a null matching pairwise *correlations* across neurons would be a still stronger structure test. (iii) Gate C randomizes one projection; joint randomization of wiring and channel is a stronger ablation. (iv) The 6-seed battery bounds the exact paired permutation resolution at $1/64$; we report the $n = 20$ seed-level test alongside it and keep per-seed reporting honest (Hurlbert, 1984). (v) All results are for this hub at this operating point; the protocol transfers, the numbers do not.

5.5 Future work

Two directions follow directly: (a) a *structure*-matching null with pairwise correlations, to test substrate *structure* rather than rates; (b) replaying the *real* closed-loop data through the same gates, converting this open-loop instrument into a closed-loop falsification battery (Carmena et al., 2003; Zhou et al., 2026). Gate G covers one alternative architecture; a broader sweep (connectivity topology, degree distribution, decision-map shape) would test the ranking further.

6 Reproducibility and data

- Code: `sandbox/` git repo; Gates B/B2/C/D/D bis/E/F/G runners (`gate_b/run.py`, `04_p2/gate_b2/run3.py`, `gate_d/run.py`, `gate_d/run_mlp.py`, `gate_d/run_20seeds.py`,

`gate_d/run_archB.py`, `gate_e/run.py`) print JSON summaries into their `results/`; tables/figures are *generated* from those JSON artifacts by `build_figures_tables.py` (no hand-typed numbers).

- Gate C: `verify/verify_wr.py` (per-seed random readouts, RNG 4242) writes `verify_wr_summary.json`.
- Gate D at $n = 20$: `gate_d/run_20seeds.py` writes `gate_d/results20/gate_d_summary.json` and `gate_d20_seed_stats.json`; the 6-seed artifacts are untouched. Gate D bis: `gate_d/run_mlp.py` writes `gate_d/results/mlp/mlp_summary.json`. Gate G (architecture B): `gate_d/run_archB.py` patches the hub geometry (`N`, `p_re`) and writes `gate_d/results_archB/`; the architecture A artifacts are untouched.
- Number audit: `verify/verify_paper_numbers.py` re-derives the quoted values from the JSON artifacts and exits non-zero on any mismatch.
- Environment: Python 3.12, numpy 2.5, matplotlib; venv at `02_c1/.venv312/`; `requirements.txt` pins versions.
- Scope: this note's claims rest on Gates B/B2/C/D/D bis/E/F/G only (all artifacts above). A companion closed-loop battery lives in the separate `surrogate_c1/` repository; it is *not* regenerated here and none of the results in this note depend on it (Zhou et al., 2026).
- Build note: this v2 compiled with MiKTeX ('pdflatex'); no Overleaf-only packages.

References

- Balci, F., Ben Hamed, S., Boraud, T., Bouret, S., Brochier, T., Brun, C., others. (2023). A response to claims of emergent intelligence and sentience in a dish. *Neuron*, 111(5), 604-605.
- Bartolozzi, C., Indiveri, G., Donati, E. (2022). Embodied neuromorphic intelligence. *Nature Communications*, 13, 1024.
- Brooks, R. A. (1991). Intelligence without representation. *Artificial Intelligence*, 47, 139-159.
- Carmena, J. M., others. (2003). Learning to control a brain-machine interface for reaching and grasping by primates. *PLoS Biology*, 1(2), e42.
- D’Angelo, G., Pedersen, J. E., Hassan, T., others. (2026). A benchmarking framework for embodied neuromorphic agents. *Nature Machine Intelligence*, 8, 300-312.
- Ernst, M. D. (2004). Permutation methods: A basis for exact inference. *Statistical Science*, 19(4), 676-685.
- Georgopoulos, A. P., Schwartz, A. B., Kettner, R. E. (1986). Neuronal population coding of movement direction. *Science*, 233, 1416-1419.
- Gerstner, W., Kistler, W. M. (2002). *Spiking neuron models: Single neurons, populations, plasticity*. Cambridge University Press.
- Hoerl, A. E., Kennard, R. W. (1970). Ridge regression: Biased estimation for nonorthogonal problems. *Technometrics*, 12, 55-67.
- Hurlbert, S. H. (1984). Pseudoreplication and the design of ecological field exper-

- iments. *Ecological Monographs*, 54(2), 187-211.
- Izhikevich, E. M. (2003). Simple model of spiking neurons. *IEEE Transactions on Neural Networks*, 14(6), 1569-1572.
- Izhikevich, E. M. (2004). Which model to use for cortical spiking neurons? *IEEE Transactions on Neural Networks*, 15(5), 1063-1070.
- Jaeger, H. (2001). The echo state approach to analysing and training recurrent neural networks (GMD-Report 148). GMD.
- Kagan, B. J., others. (2023). Scientific communication and the semantics of sentience. *Neuron*, 111(5). [Letter].
- Kagan, B. J., Kitchen, A. C., Tran, N. T., Habibollahi, F., Khajehnejad, M., Parker, B. J., Bhat, A., Rollo, B., Razi, A., Friston, K. J. (2022). In vitro neurons learn and exhibit sentience when embodied in a simulated game-world. *Neuron*, 110(22), 3952-3969.
- Kraskov, A., Stoegbauer, H., Grassberger, P. (2004). Estimating mutual information. *Physical Review E*, 69, 066138.
- Lukosevicius, M., Jaeger, H. (2009). Reservoir computing approaches to recurrent neural network training. *Computer Science Review*, 3(3), 127-149.
- Maass, W. (1997). Networks of spiking neurons: The third generation of neural network models. *Neural Networks*, 10(9), 1659-1671.
- Maass, W., Natschlaeger, T., Markram, H. (2002). Real-time computing without stable states: A new framework for neural computation based on perturbations. *Neural Computation*, 14(11), 2531-2560.

- Nicolelis, M. A. L. (2003). Brain-machine interfaces to restore motor function and probe neural circuits. *Nature Reviews Neuroscience*, 4, 417-422.
- Paninski, L., others. (2004). Superlinear population encoding of dynamic hand trajectory in primary motor cortex. *Journal of Neuroscience*, 24(39), 8551-8561.
- Panzeri, S., Senatore, R., Montemurro, M. A., Petersen, R. S. (2007). Correcting for the sampling bias in spike-train information measures. *Journal of Neurophysiology*, 98, 1064-1072.
- Pfeifer, R., Bongard, J. (2006). How the body shapes the way we think: A new view of intelligence. MIT Press.
- Quiroga, R. Q., Panzeri, S. (2009). Extracting information from neuronal populations: Information theory and decoding approaches. *Nature Reviews Neuroscience*, 10, 173-185.
- Schreiber, T. (2000). Measuring information transfer. *Physical Review Letters*, 85(2), 461-464.
- Shannon, C. E. (1948). A mathematical theory of communication. *Bell System Technical Journal*, 27, 379-423.
- Smirnova, L., Caffo, B. S., Gracias, D. H., Huang, Q., Morales Pantoja, I. E., others. (2023). Organoid intelligence (OI): The new frontier in biocomputing and intelligence-in-a-dish. *Frontiers in Science*, 1, 1017235.
- Steinmetz, N. A., Zatka-Haas, P., Carandini, M., Harris, K. D. (2019). Distributed coding of choice, action and engagement across the mouse brain. *Nature*, 576, 266-273.

- Theiler, J., Eubank, S., Longtin, A., Galdrikian, B., Farmer, J. D. (1992). Testing for nonlinearity in time series: The method of surrogate data. *Physica D*, 58, 77-94.
- Walker, B. L., Newhall, K. A. (2018). Inferring information flow in spike-train data sets using a trial-shuffle method. *PLoS ONE*, 13(11), e0206977.
- Wringe, C., Trefzer, M., Stepney, S. (2025). Reservoir computing benchmarks: A tutorial review and critique. *International Journal of Parallel, Emergent and Distributed Systems*, 40(4), 313-351.
- Yan, M., Huang, C., Bienstman, P., Tiño, P., Lin, W., Sun, J. (2024). Emerging opportunities and challenges for the future of reservoir computing. *Nature Communications*, 15, 2056.
- Yik, J., Van den Berghe, K., den Blanken, D., others. (2025). The NeuroBench framework for benchmarking neuromorphic computing algorithms and systems. *Nature Communications*, 16, 1545.
- Zhou, J., Tanneberg, D., Habibollahi, F., Loeffler, A., Lawson, K., Baccetti, V., others. (2026). Embodied neurocomputation: A framework for interfacing biological neural cultures with scaled task-driven validation. [Preprint]. arXiv:2605.13315.